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Sem1 BCA sec A

Assessment task 2

AI vs ML vs DL: the nested circles, one at a time.

Q1) What is AI?

AI (Artificial Intelligence) is the big umbrella idea: making machines do things that normally need human intelligence, like understanding language, recognizing faces, making decisions, or playing chess.

Q2) What is ML?

ML (Machine Learning) is a way of achieving AI. Instead of writing exact rules for the computer, you show it lots of examples and it learns the patterns itself. For instance, show it thousands of emails labelled "spam" or "not spam," and it learns to spot spam on its own.

Q3) What is DL?

DL (Deep Learning) is a type of ML that uses many-layered neural networks, loosely inspired by how brain cells connect. It's especially good with messy, complex data like images, speech, and text. Face lock, voice assistants, and chatbots all use it.

Q4) Separate a rule from ML

System A is a rule (hand-written logic).

System B is ML.

System A: rule. A programmer wrote the exact condition: "if the required box is empty, block the form." The behaviour is fully decided in advance, it never changes unless someone edits the code, and the same input always gives the same output for the same reason.

System B: ML. No one wrote "after this answer, show question 14." Instead, the system learns patterns from past learners' answers and uses them to predict what a learner is likely to need next. Its behaviour comes from data, and it can improve or change as more data arrives.

Q5) Separate ML from GenAI

Tool 1 (predicts which existing film a viewer may like): ML. It learns from past viewing data and picks from films that already exist. It predicts or selects; it doesn't create anything new.

Tool 2 (writes a new two-line film description): Gen AI. It produces brand-new content (text) that didn't exist before, rather than choosing from a fixed list.

Q6) Compare two mechanisms

Part 1: Fixed five-attempt login lock: Rule (exact, hand-written logic). A programmer wrote the exact condition ("after five failed attempts, lock"), so the behaviour is decided in advance.

Part 2: Face check (edges to shapes through stacked layers): Deep Learning. Many layers of a neural network learn features on their own, starting with simple edges and building up to shapes and then a face.

Part 3: Mechanism difference: the login lock follows a condition a human wrote explicitly, while the face check works out its own internal features from training data through layered learning, with no human writing the matching steps.

Q7) Correct two mixed-up claims

"Anything that responds quickly is ML." Wrong. Speed doesn't decide the category. Use the lesson test: *did a human write the exact decision condition, or did the system work it out from examples?* A rule can respond instantly because it just checks a condition, and an ML system can be slow. How the decision is made is what counts, not how fast.

"DL means smarter AI." Wrong. DL is not a higher level of intelligence. It is one kind of ML that uses many stacked layers to learn features on its own, building from simple edges to shapes to whole objects. It suits messy data like images and speech, but a simple rule can be the better, more reliable choice for a simple task.

Examples:

- **Exact rule (fast but not ML):** "If the required box is empty, block the form." A programmer wrote this exact condition, and it responds instantly.
- **Layered DL:** A face check that learns edges, then shapes, then a face through stacked layers. It is a better fit for faces than a rule, not "smarter" in general.